Chapter 12: Aggregate Demand and Aggregate Supply

Practice Problems

Warm-Up: Quick Concept and Calculation Checks

a. For each scenario, name which of the three reasons the AD curve slopes downward is at work:
   (i) a higher price level erodes the purchasing power of household savings accounts, so people spend less; (ii) a higher price level makes domestic goods pricier than foreign alternatives, reducing exports; (iii) a higher price level increases the demand for money, pushing up interest rates and discouraging interest-sensitive investment.

b. Is the short-run aggregate supply curve (SRAS) upward sloping or vertical? What about the long-run aggregate supply curve (LRAS)?

c. Which curve, SRAS or LRAS, reflects the idea that input prices (especially wages) are sticky and have not yet adjusted?

d. A new technology raises productivity across the whole economy. Does this shift AS right or left?

e. A key imported input suddenly becomes much more expensive. Does this shift AS right or left?

f. True or False: an increase in AD moves the price level and real GDP in the *same* direction, while a decrease in AS (a leftward shift) also raises the price level but *lowers* real GDP.

Problem 1: Sizing an AD Shift with the Multiplier

A country's MPC is 0.8. A wave of business optimism leads firms to increase planned investment spending by $45 billion at every price level.

a. What is the multiplier?

b. By how much does this initial change in investment ultimately shift the AD curve (measured in real GDP at a given price level)?

c. Name one other component of aggregate demand (besides investment) that could shift AD the same way, and briefly explain why.

d. If MPC had instead been 0.5, would the same $45 billion increase in investment shift AD by *more* or *less* than in part (b)? Explain in one sentence.

Problem 2: Demand-Pull vs. Cost-Push Inflation

For each event below, determine whether it causes *demand-pull* or *cost-push* inflation, and predict what happens to real GDP alongside the price level.

a. Consumer confidence surges and households sharply increase spending at every price level.

b. A major hurricane disrupts oil refineries, sharply raising energy costs for nearly every firm in the economy.

c. The government significantly increases spending on infrastructure projects.

d. A new tax is imposed on a key raw material used throughout manufacturing.